



What the Raised Hands Don't Say

A Survey of Comprehension Checks Against the Device-Free Hour Hand-Raise Tally

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The problem

Slop University's Device-Free Hour pilot renews its continuation case each term on one figure: how many hands go up. Whether a rising tally tracks what students can actually show they understood, rather than how often they reach for the ceiling, has not been checked.

Research questions

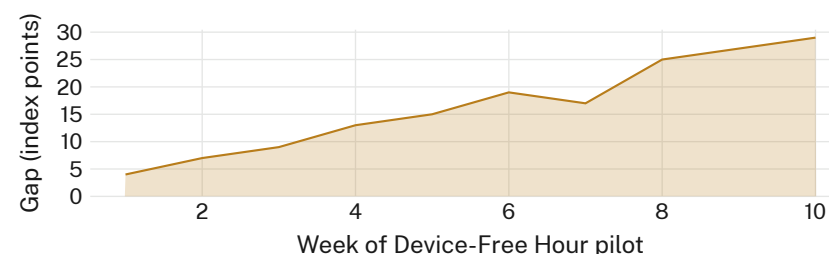
- Do session hand-raise counts predict comprehension-check pass rates?
- Do lecturing staff and students read the same session the same way?
- Does the tally-comprehension gap hold steady or widen across the pilot?

Study design

- n = 214 first-year students · 16 sessions · 4 courses · one teaching period
- Paired per session: TA-logged hand-raise tally · blind-scored 5-item comprehension check
- Parallel staff engagement-perception survey (n = 7 lecturing staff); scoring thresholds pre-registered

What we found

Session hand-raise counts explain 12% of the variance in comprehension-check pass rates ($r = 0.34$, $n = 214$). Staff engagement ratings track the tally ($r = 0.68$) more closely than either measure tracks comprehension.



The gap between the session hand-raise index and the comprehension-check pass rate widened from 4 points in week 1 to 29 points by week 10, tracking the tally's own rise almost exactly.

Implications & where next

The continuation case reads the tally as a stand-in for understanding; the comprehension checks say the two parted ways by week 4 and kept drifting. Staff ratings move with the raised hands, not the quiz. Next: pair the tally with a comprehension check by default, not as a one-term add-on.

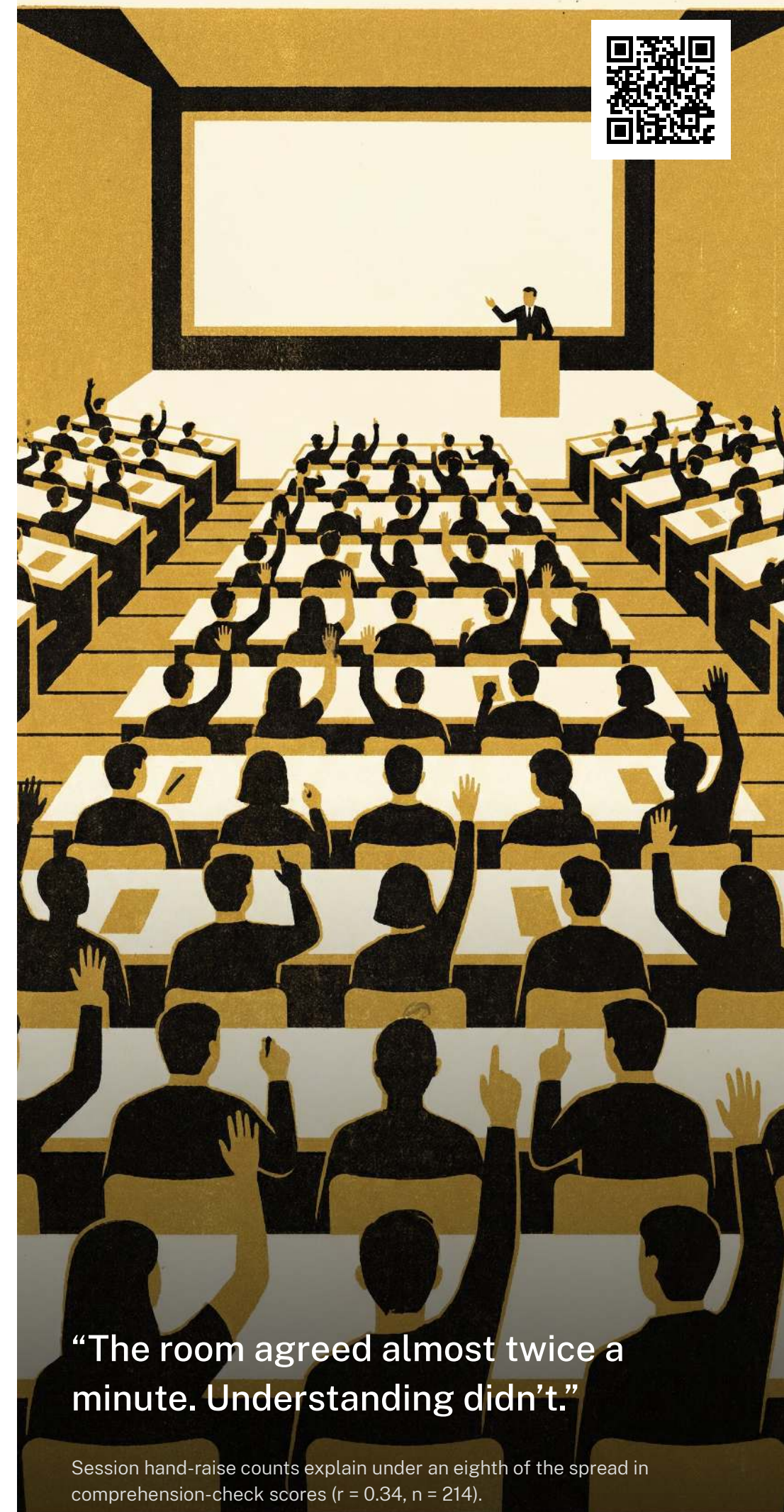
Selected references

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Comprehension checks scored blind to each session's hand-raise count; results reported in aggregate only.

We thank the lecturing staff who let a tally-keeper sit in on sessions already running device-free.

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“The room agreed almost twice a minute. Understanding didn't.”

Session hand-raise counts explain under an eighth of the spread in comprehension-check scores ($r = 0.34$, $n = 214$).